



Prince of Songkla University

Phuket Campus

SCI-104 – Mathematical Foundations and Quantum Mechanics Essentials

Lab 1 Report: Qubits and Quantum States

Lab	Lab 1 – Qubit, Dirac Notation, and the Bloch Sphere
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Abstract

In this lab, single-qubit states were represented in Dirac notation and visualized on the Bloch sphere using Qiskit. The basis states $|0\rangle$, $|1\rangle$, the superposition states $|+\rangle$, $|-\rangle$, and a state produced by $R_y(\pi/3)$ were built as quantum circuits, and their state vectors and Bloch vectors were compared with hand calculations. All simulated results agreed with theory (for example, $R_y(\pi/3)|0\rangle = 0.866|0\rangle + 0.5|1\rangle$ with $P(0) = 0.75$). The exercises showed that states with $\theta = \pi/2$ lie on the equator, that a global phase changes neither probabilities nor the Bloch vector, and that R_x , R_y , R_z rotate the Bloch vector about the x , y , z axes. The main lesson is that the relative phase, not the global phase, distinguishes states such as $|+\rangle$ and $|-\rangle$ that have identical measurement probabilities.

1. Objectives

1. To represent a qubit in **Dirac notation** and in the computational basis.
2. To translate between Dirac notation and column-vector form.
3. To use **Qiskit** to build and visualize single-qubit states.
4. To plot and interpret **Bloch sphere** representations.
5. To observe how quantum gates (X , H , R_x , R_y , R_z) move states on the Bloch sphere.

2. Background

2.1 The Qubit

A classical bit is either 0 or 1. A qubit can be in a **superposition** of both basis states, described by two complex numbers called amplitudes. When the qubit is measured, the result is 0 or 1 with probabilities given by the squared magnitudes of the amplitudes.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

2.2 Dirac Notation

Symbol	Name	Meaning
$ \psi\rangle$	ket	A state written as a column vector, e.g. $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
$\langle\psi $	bra	The conjugate transpose of the ket, a row vector $[\alpha^* \ \beta^*]$
$\langle\phi \psi\rangle$	inner product	A complex number measuring the overlap of two states; $\langle\psi \psi\rangle = 1$ for a normalized state
$ \alpha ^2$	—	Probability of measuring 0
$ \beta ^2$	—	Probability of measuring 1

2.3 Bloch Sphere Representation

Any pure single-qubit state can be written, up to a global phase, as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi < 2\pi$$

Cartesian coordinates of the Bloch vector:

$$(x, y, z) = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$$

θ is the polar angle measured from the $+z$ axis and φ is the azimuthal angle around the z axis. The north pole ($\theta = 0$) is $|0\rangle$ and the south pole ($\theta = \pi$) is $|1\rangle$. The equator ($\theta = \pi/2$) holds the equal-probability superpositions, with φ choosing the direction on the equator. The half-angle $\theta/2$ appears because $|0\rangle$ and $|1\rangle$ are orthogonal (90°) as vectors but opposite (180°) on the sphere.

2.4 Gates Used in This Lab

Gate	Matrix	Effect on the Bloch sphere
X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Rotation by π about x ; swaps $ 0\rangle \leftrightarrow 1\rangle$ (north $\leftrightarrow$ south pole)
H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Rotation by π about the $(x+z)/\sqrt{2}$ axis; $ 0\rangle \rightarrow +\rangle$ ($+z \rightarrow +x$)
$R_x(\lambda)$	$\begin{bmatrix} \cos \frac{\lambda}{2} & -i \sin \frac{\lambda}{2} \\ -i \sin \frac{\lambda}{2} & \cos \frac{\lambda}{2} \end{bmatrix}$	Rotates the Bloch vector by λ about the x axis
$R_y(\lambda)$	$\begin{bmatrix} \cos \frac{\lambda}{2} & -\sin \frac{\lambda}{2} \\ \sin \frac{\lambda}{2} & \cos \frac{\lambda}{2} \end{bmatrix}$	Rotates the Bloch vector by λ about the y axis

Gate	Matrix	Effect on the Bloch sphere
$R_z(\lambda)$	$\begin{bmatrix} e^{-i\lambda/2} & 0 \\ 0 & e^{i\lambda/2} \end{bmatrix}$	Rotates the Bloch vector by λ about the z axis (changes φ)

Note: λ is the rotation angle of the gate. It is different from the Bloch sphere angle θ in Section 2.3.

3. Tools and Environment

Item	Version / Detail
Platform	[Google Colab / local machine — edit to match where you ran it] (results below were generated on a local macOS machine)
Python	3.9.6
qiskit	2.2.3
Other libraries	numpy 2.0.2, matplotlib 3.9.4, pylatexenc 2.11

4. Lab Tasks and Results

Task 1 — Setup

```
!pip install qiskit qiskit-aer
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_bloch_multivector
```

Observation: Qiskit imported successfully. The package `pylatexenc`, which some Qiskit drawing styles need, was missing and was installed with `pip install pylatexenc`. A

`DeprecationWarning` states that Qiskit will drop Python 3.9 support in version 2.3.0; it does not affect the results.

Task 2 — Representing $|0\rangle$ and $|1\rangle$

```
qc0 = QuantumCircuit(1)
qc1 = QuantumCircuit(1); qc1.x(0)

state0 = Statevector.from_instruction(qc0)
state1 = Statevector.from_instruction(qc1)

display(plot_bloch_multivector(state0))
display(plot_bloch_multivector(state1))
```

State	Statevector output	Bloch sphere
$ 0\rangle$	<code>[1.+0.j, 0.+0.j]</code>	A Bloch sphere diagram for qubit 0. The sphere is centered at the origin of a 3D coordinate system with axes labeled x, y, and z. The north pole is labeled $0\rangle$ and the south pole is labeled $1\rangle$. A vertical pink arrow points from the center to the north pole, representing the Bloch vector for the state $0\rangle$.
$ 1\rangle$	<code>[0.+0.j, 1.+0.j]</code>	A Bloch sphere diagram for qubit 0. The sphere is centered at the origin of a 3D coordinate system with axes labeled x, y, and z. The north pole is labeled $0\rangle$ and the south pole is labeled $1\rangle$. A vertical pink arrow points from the center to the south pole, representing the Bloch vector for the state $1\rangle$.

Observation: $|0\rangle$ has Bloch vector $(0, 0, 1)$ and points to the north pole ($\theta = 0$). $|1\rangle$ has Bloch vector $(0, 0, -1)$ and points to the south pole ($\theta = \pi$). The X gate moved the state from one pole to the other. Measuring gives $P(0) = 1$ for $|0\rangle$ and $P(1) = 1$ for $|1\rangle$.

Task 3 — Superposition States

```
qc_plus = QuantumCircuit(1); qc_plus.h(0)
qc_minus = QuantumCircuit(1); qc_minus.x(0); qc_minus.h(0)

state_plus = Statevector.from_instruction(qc_plus)
state_minus = Statevector.from_instruction(qc_minus)

display(plot_bloch_multivector(state_plus))
display(plot_bloch_multivector(state_minus))
```

State	Dirac form (by hand)	Statevector output	Bloch sphere
$ +\rangle$	$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	<code>[0.7071+0.j, 0.7071+0.j]</code>	A Bloch sphere diagram for qubit 0. The vertical axis is labeled 0> at the top and 1> at the bottom. The horizontal axis is labeled x on the left and y on the right. A pink arrow points from the center to the surface at approximately 45 degrees from the 0> axis in the x-z plane.
$ -\rangle$	$\frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	<code>[0.7071+0.j, -0.7071+0.j]</code>	A Bloch sphere diagram for qubit 0. The vertical axis is labeled 0> at the top and 1> at the bottom. The horizontal axis is labeled x on the left and y on the right. A pink arrow points from the center to the surface at approximately 45 degrees from the 0> axis in the y-z plane.

Hand calculation:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = |-\rangle$$

($|-\rangle$ was built as $HX|0\rangle = H|1\rangle$.)

Expected vs. observed:

State	Expected position	Observed position	Match?
$ +\rangle$	equator, $+x$ axis	Bloch vector $(1, 0, 0)$	✓
$ -\rangle$	equator, $-x$ axis	Bloch vector $(-1, 0, 0)$	✓

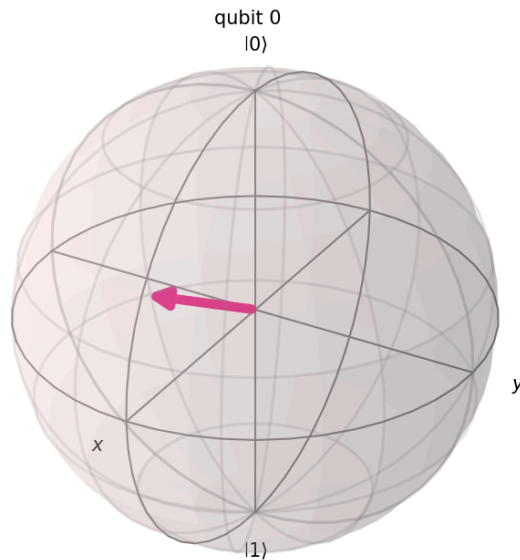
Both states have $P(0) = P(1) = 0.5$, so a measurement cannot tell them apart. They differ only in the sign (relative phase $\varphi = \pi$) of the $|1\rangle$ amplitude, which appears as opposite directions on the x axis.

Task 4 — Arbitrary Rotation

```
from math import pi
qc = QuantumCircuit(1)
qc.ry(pi/3, 0) # Rotate around Y by 60 degrees
state = Statevector.from_instruction(qc)
plot_bloch_multivector(state)
```

Hand calculation: $R_y(\lambda)|0\rangle = \cos \frac{\lambda}{2}|0\rangle + \sin \frac{\lambda}{2}|1\rangle$, so with $\lambda = \pi/3$: $\alpha = \cos 30^\circ = \frac{\sqrt{3}}{2}$, $\beta = \sin 30^\circ = \frac{1}{2}$.

Quantity	Hand calculation	Qiskit result
α	$\frac{\sqrt{3}}{2} \approx 0.866$	0.866
β	$\frac{1}{2} = 0.5$	0.5
$P(0) = \alpha ^2$	$\frac{3}{4} = 0.75$	0.75
$P(1) = \beta ^2$	$\frac{1}{4} = 0.25$	0.25
(θ, φ)	$(\pi/3, 0)$	Bloch vector $(0.866, 0, 0.5)$, i.e. $\theta = 60^\circ$, $\varphi = 0$



Observation: Starting from the north pole, the state moved 60° toward $+x$ in the xz -plane. The Bloch vector $(\sin 60^\circ, 0, \cos 60^\circ) = (0.866, 0, 0.5)$ agrees exactly with the hand calculation. Since α and β are real, $\varphi = 0$.

5. Exercises

Exercise 1 — General State Exploration ($\theta = \pi/4$, $\varphi = \pi/2$)

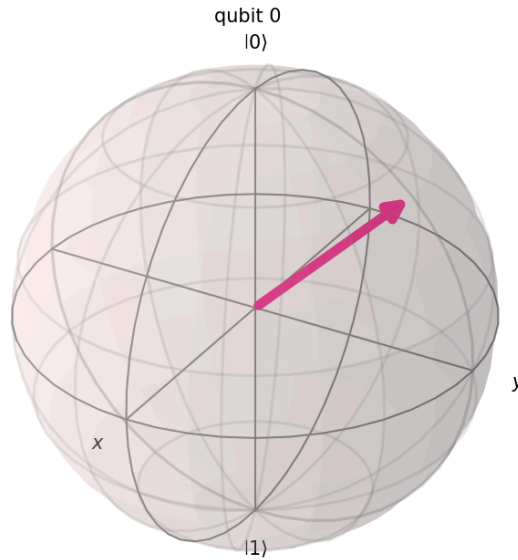
Dirac form (by hand):

$$|\psi\rangle = \cos \frac{\pi}{8} |0\rangle + e^{i\pi/2} \sin \frac{\pi}{8} |1\rangle = 0.924 |0\rangle + 0.383 i |1\rangle$$

Expected Bloch vector: $(\sin \frac{\pi}{4} \cos \frac{\pi}{2}, \sin \frac{\pi}{4} \sin \frac{\pi}{2}, \cos \frac{\pi}{4}) = (0, 0.707, 0.707)$.

Code:

```
qc = QuantumCircuit(1)
qc.ry(pi/4, 0) # sets theta
qc.rz(pi/2, 0) # sets phi
state = Statevector.from_instruction(qc)
print(state)
display(plot_bloch_multivector(state))
```



Observation: Qiskit gave `[0.6533-0.6533j, 0.2706+0.2706j]` with $P(0) = 0.854$, $P(1) = 0.146$ and Bloch vector $(0, 0.707, 0.707)$, matching the expected values. The amplitudes differ from my hand form by a factor $e^{-i\pi/4}$: Qiskit's $R_z(\lambda) = \text{diag}(e^{-i\lambda/2}, e^{i\lambda/2})$ includes a global phase. A global phase does not change the physical state (see Exercise 3), and the relative phase of $|1\rangle$ to $|0\rangle$ is still $e^{i\pi/2}$.

Exercise 2 — Equator States

Proof / argument: For $\theta = \pi/2$, the Bloch vector is $(\sin \frac{\pi}{2} \cos \varphi, \sin \frac{\pi}{2} \sin \varphi, \cos \frac{\pi}{2}) = (\cos \varphi, \sin \varphi, 0)$. Its z component is 0 for every φ , so the state lies in the xy -plane, which is the equator. The probabilities are $\cos^2 \frac{\pi}{4} = \sin^2 \frac{\pi}{4} = \frac{1}{2}$, independent of φ .

Code (test several values of φ):

```
for phi in [0, pi/2, pi, 3*pi/2]:
    qc = QuantumCircuit(1)
    qc.ry(pi/2, 0)
    qc.rz(phi, 0)
    state = Statevector.from_instruction(qc)
    display(plot_bloch_multivector(state))
```

φ	Bloch vector (x, y, z)	On equator?
0	(1, 0, 0)	✓
$\pi/2$	(0, 1, 0)	✓
π	(-1, 0, 0)	✓

φ	Bloch vector (x, y, z)	On equator?
$3\pi/2$	$(0, -1, 0)$	✓

All four points have $z = 0$ and lie 90° apart around the equator, and every one gives $P(0) = P(1) = 0.5$.

Exercise 3 — Global Phase

Proof: Let $|\psi'\rangle = e^{i\gamma}|\psi\rangle = e^{i\gamma}\alpha|0\rangle + e^{i\gamma}\beta|1\rangle$. Then

$$|e^{i\gamma}\alpha|^2 = |e^{i\gamma}|^2|\alpha|^2 = |\alpha|^2,$$

and likewise for β , so the probabilities are unchanged. The relative phase is also unchanged, since $\frac{e^{i\gamma}\beta}{e^{i\gamma}\alpha} = \frac{\beta}{\alpha}$, so θ and φ , and hence the Bloch vector, are unchanged.

Code:

```
import numpy as np
sv = Statevector.from_instruction(qc)          # state from Exercise 1
sv2 = Statevector(sv.data * np.exp(1j * 1.234)) # global phase gamma = 1.234
print(sv.probabilities(), sv2.probabilities())
display(plot_bloch_multivector(sv)); display(plot_bloch_multivector(sv2))
```

Result: Both states give probabilities `[0.8536, 0.1464]` and the same Bloch vector $(0, 0.707, 0.707)$. The two Bloch spheres are identical, confirming that a global phase is not physically observable.

Exercise 4 — Rotation Experiment

Code:

```
for gate in ["rx", "ry", "rz"]:
    qc = QuantumCircuit(1)
    getattr(qc, gate)(pi/2, 0)
    state = Statevector.from_instruction(qc)
    print(gate, state)
    display(plot_bloch_multivector(state))
```

Gate applied to $ 0\rangle$	Statevector	Final Bloch vector (x, y, z)	Bloch sphere
$R_x(\pi/2)$	<code>[0.7071+0.j, 0-0.7071j]</code>	$(0, -1, 0)$	
$R_y(\pi/2)$	<code>[0.7071+0.j, 0.7071+0.j]</code>	$(1, 0, 0)$	
$R_z(\pi/2)$	<code>[0.7071-0.7071j, 0+0j]</code>	$(0, 0, 1)$	

Observation: $R_x(\pi/2)$ rotates the north pole by 90° about x and ends at $-y$. $R_y(\pi/2)$ rotates it about y and ends at $+x$. $R_z(\pi/2)$ leaves the state at the north pole, because $|0\rangle$ already lies on the z axis, the rotation axis. Its only effect is a global phase $e^{-i\pi/4}$, which is unobservable. This shows that R_z only changes φ and cannot move a state that sits on the z axis.

6. Concept Questions

Q1. Geometrically, what does $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ represent on the Bloch sphere?

This is the state $|+\rangle$. It has $\theta = \pi/2$ and $\varphi = 0$, so it lies on the equator at the $+x$ axis, Bloch vector $(1, 0, 0)$. It is an equal superposition with $P(0) = P(1) = 0.5$.

Q2. Why must $|\alpha|^2 + |\beta|^2 = 1$ for a valid qubit state?

$|\alpha|^2$ and $|\beta|^2$ are the probabilities of the two possible measurement outcomes, and the probabilities of all outcomes must sum to 1. Equivalently, a state must be a unit vector ($\langle\psi|\psi\rangle = 1$), which is also what lets it be placed on the surface of the unit Bloch sphere.

Q3. What is the difference between a global phase and a relative phase?

A global phase multiplies the whole state, $e^{i\gamma}(\alpha|0\rangle + \beta|1\rangle)$. It changes neither the probabilities nor the Bloch vector, so it has no physical effect. A relative phase is the phase difference between the $|0\rangle$ and $|1\rangle$ amplitudes (φ). It does change the state: it moves the Bloch vector around the z axis and, for example, distinguishes $|+\rangle$ from $|-\rangle$.

Q4. Which states lie at the poles of the Bloch sphere?

The north pole ($\theta = 0$) is $|0\rangle$ and the south pole ($\theta = \pi$) is $|1\rangle$. These are the computational basis states and are the definite outcomes of a measurement in the z basis.

Q5. How do the rotation gates R_x , R_y , R_z correspond to movements on the Bloch sphere?

$R_x(\lambda)$, $R_y(\lambda)$ and $R_z(\lambda)$ rotate the Bloch vector by the angle λ about the x , y and z axes respectively. R_z changes only the azimuthal angle φ (relative phase), while R_x and R_y change θ and so change the measurement probabilities. Every single-qubit gate corresponds to some rotation of the sphere.

7. Discussion

Theory vs. simulation. All Qiskit results matched the hand calculations: the state vectors of $|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$, the amplitudes 0.866 and 0.5 for $R_y(\pi/3)|0\rangle$, and every Bloch vector in Exercises 2 and 4. The only difference appeared in Exercise 1, where Qiskit's amplitudes carried an extra global phase $e^{-i\pi/4}$ because its R_z is defined as $\text{diag}(e^{-i\lambda/2}, e^{i\lambda/2})$. This did not change the physical state, which matches the result of Exercise 3.

Problems faced. The `pylatexenc` package was missing and had to be installed. When the script was first saved inside a folder named `qiskit` (a clone of the Qiskit source code), Python imported that folder instead of the installed package and failed with a circular-import error; moving the script to another folder fixed it. In a terminal the Bloch spheres cannot be displayed, so they were saved as PNG files, while a notebook can show them directly with `display(...)`.

Key insight. The most important point is that probabilities do not describe a state completely. $|+\rangle$ and $|-\rangle$ give identical measurement statistics in the computational basis but are different states, separated by a relative phase of π . The Bloch sphere makes this visible, since the two states sit at opposite ends of the x axis, and it shows that global phase is the only part of the amplitudes that carries no physical meaning.

8. Conclusion

In this lab, single-qubit states were written in Dirac notation and visualized on the Bloch sphere with Qiskit. Basis states appeared at the poles, $|\pm\rangle$ on the equator along the x axis, and $R_y(\pi/3)|0\rangle$ at 60° from the north pole, all in agreement with hand calculations. The exercises confirmed that equator states have $\theta = \pi/2$, that global phase is unobservable, and that R_x , R_y , R_z are rotations about the corresponding axes. All the objectives of the lab were met.

9. References

1. IBM Quantum Learning (formerly the Qiskit Textbook).
<https://quantum.cloud.ibm.com/learning>
2. M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.
3. J. Preskill, *Lecture Notes for Physics 229: Quantum Computation*, Caltech.
4. IBM Quantum Documentation. <https://quantum.cloud.ibm.com/docs>
5. Lab sheet: komsan-k, *Lab 1: Qubits and Quantum States*. <https://github.com/komsan-k/quantum-computation/tree/main/lab/lab-1-qubit-dirac-bloch>

Appendix — Full Code

```
import numpy as np
from math import pi
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_bloch_multivector

def run(build):
    qc = QuantumCircuit(1); build(qc)
    sv = Statevector.from_instruction(qc)
    print(np.round(sv.data, 4), np.round(sv.proBABILITIES(), 4))
    display(plot_bloch_multivector(sv))
    return sv

run(lambda q: None) # |0>
run(lambda q: q.x(0)) # |1>
run(lambda q: q.h(0)) # |+>
run(lambda q: (q.x(0), q.h(0))) # |->
run(lambda q: q.ry(pi/3, 0)) # Task 4
sv = run(lambda q: (q.ry(pi/4, 0), q.rz(pi/2, 0))) # Exercise 1
for phi in [0, pi/2, pi, 3*pi/2]: # Exercise 2
    run(lambda q: (q.ry(pi/2, 0), q.rz(phi, 0)))
sv2 = Statevector(sv.data * np.exp(1j*1.234)) # Exercise 3
print(np.allclose(sv.proBABILITIES(), sv2.proBABILITIES()))
for g in ["rx", "ry", "rz"]: # Exercise 4
    run(lambda q: getattr(q, g)(pi/2, 0))
```